SAR Lab 7 Report

Xinmo (Maoxian) Landslide Case Study, Sichuan, China

Jakub Błazenek

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1 Introduction

On 24 June 2017 a rock avalanche detached from a steep slope above Xinmo village (Maoxian County, Sichuan) and travelled roughly 2.6 km down the Songping Gully, burying 62 houses and killing more than 100 people [1]. Intrieri *et al.* analysed 45 Sentinel-1 images (9 Oct 2014–19 Jun 2017, descending track 62) with SqueeSAR and found precursory LOS deformation in the source area (up to 27 mm yr^{-1}), with acceleration from April 2017 that could have forecast failure within days of 24 June using Fukuzono's method.

This lab recreates a similar pre-event analysis using publicly available COMET-LiCSAR interferograms and MintPy SBAS, then compares radar- and optical-based methods for post-event mapping of the scar and surface change.

2 Data and Methods

The AOI covers 103.62°–103.68° E, 32.04°–32.09° N (Figure 1). From LiCSAR frame `062D_05831_131313` we downloaded 104 unwrapped interferograms before 24 June 2017 (baseline $\leq$ 72 days), converted them to a MintPy stack, and ran `smallbaselineApp.py` from `modify_network`. A stable reference pixel (y=45, x=55) was placed outside the source area. Post-event mapping used Sentinel-1 GRD VV and Sentinel-2 L2A composites from Google Earth Engine.

3 Pre-Event SBAS Results

Figure 2 shows the LOS velocity field and a displacement time series for a pixel on the upper slope (y=25, x=30). Negative velocity (-7.3 mm yr^{-1} at this pixel) indicates motion away from the satellite, consistent with the creeping source area described by Intrieri *et al.* Their SqueeSAR analysis resolved rates up to 27 mm yr^{-1} near the main scarp and distinguished accelerating, linear, and stable kinematic zones; our SBAS result is lower but confirms long-term instability on the slope. The village footprint itself was largely stable in the published study (-2 to $+2$ mm yr^{-1}), matching the idea that failure initiated higher on the slope.

The time series shows variable motion from 2014–2016, then a sustained negative trend from mid-2016. Intrieri *et al.* report tertiary creep with acceleration from 20 April 2017 and velocities of $\sim$2 mm day^{-1} in early June. Our LiCSAR stack ends on 7 June 2017, so the sharpest pre-failure

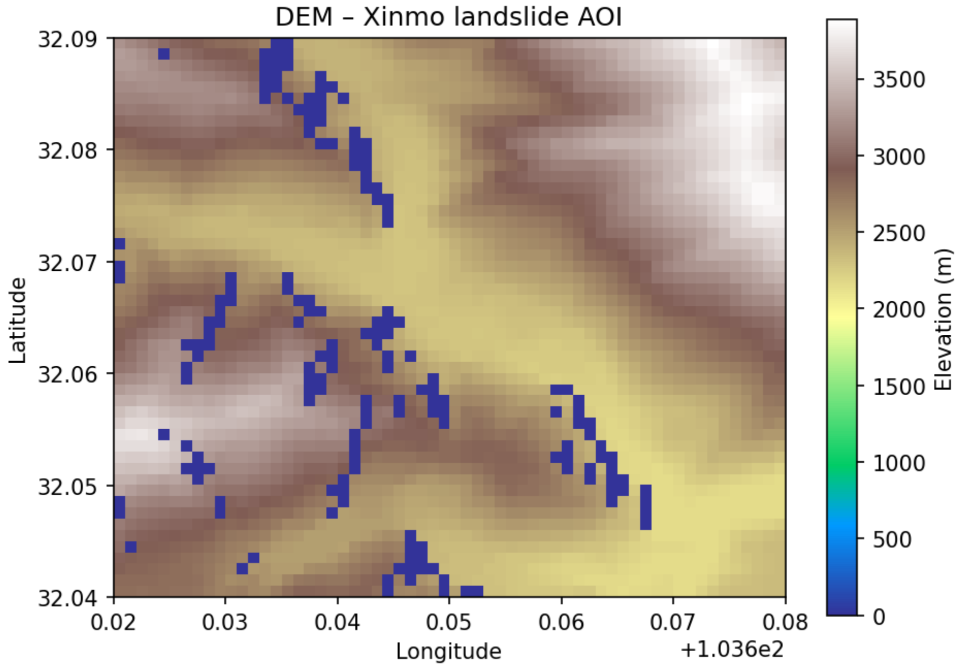
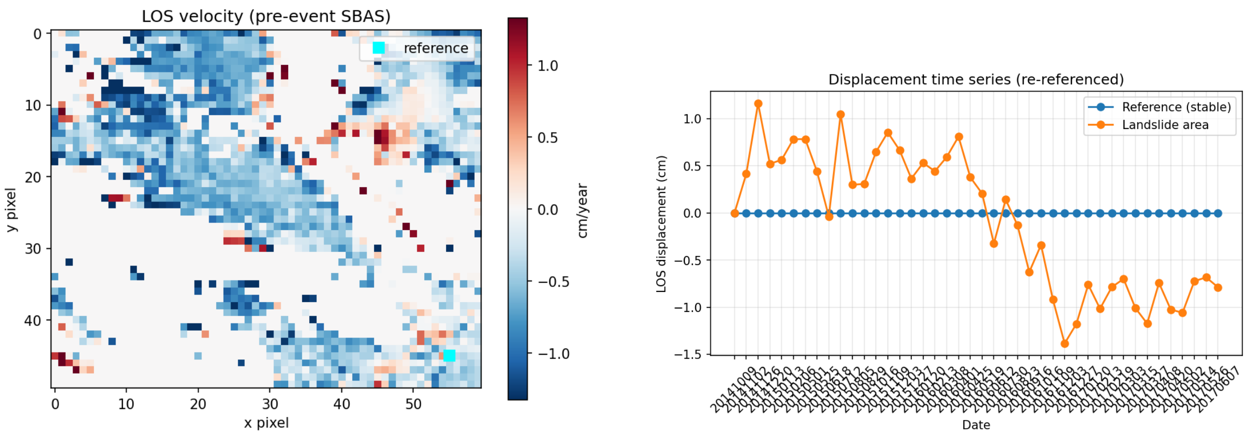


Figure 1: DEM of the Xinmo landslide AOI.

acceleration is only partly sampled—a key limitation of the pre-computed interferogram archive noted in the lab instructions.

Temporal coherence (Figure 3) is high along valley terraces but poor on steep, vegetated walls, reflecting the same decorrelation that limits InSAR point density in this terrain [1].



(a) LOS velocity (SBAS).

(b) Displacement time series.

Figure 2: Pre-event MintPy SBAS results. Cyan square: reference pixel.

4 Radar-Based Post-Event Mapping

4.1 Interferometric coherence

The LiCSAR coherence product for pair 20170526_20170607 (Figure 4) shows low values ($\bar{c} \approx 0.11$) over most of the vegetated mountains, with higher coherence only over settlements and stable benches. Widespread decorrelation prevents coherence loss from delineating the landslide

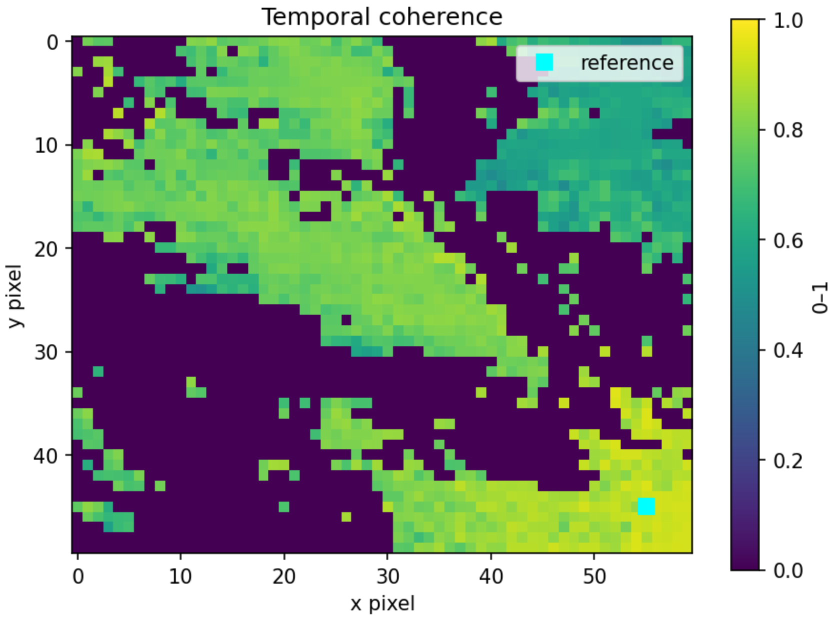


Figure 3: Temporal coherence from the SBAS inversion.

boundary—unlike the clear pre-failure deformation signal in the time-series analysis.

4.2 Sentinel-1 VV backscatter change

Pre- and post-event VV composites and their difference (Figure 5) reveal a localised anomaly along the run-out zone where debris changed surface roughness. This is more informative than coherence alone but still affected by layover on the steep slopes. VV change is therefore a **better** post-event radar mapper than coherence, but **worse** than optical data.

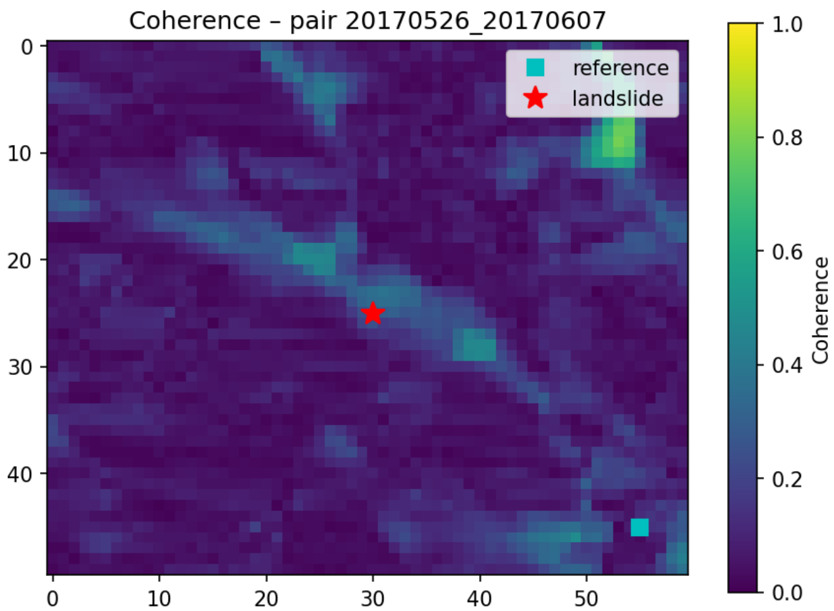
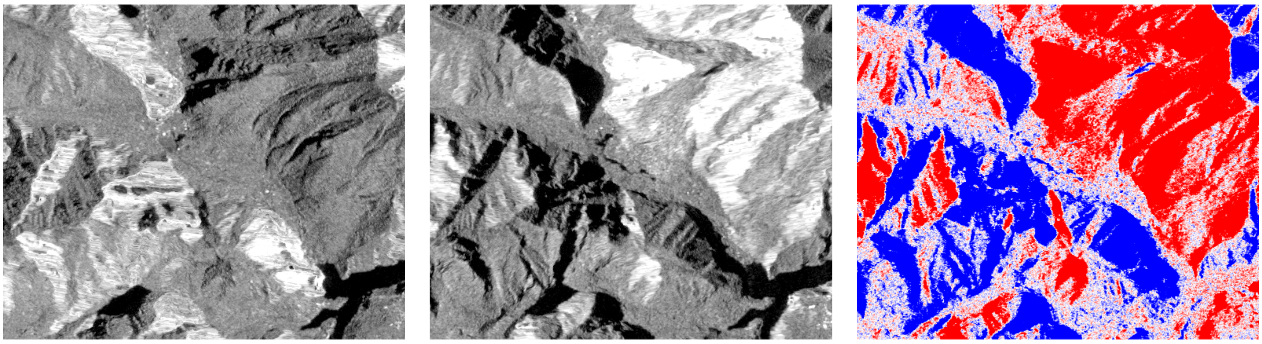


Figure 4: Interferometric coherence, pair 20170526_20170607 (AOI subset).

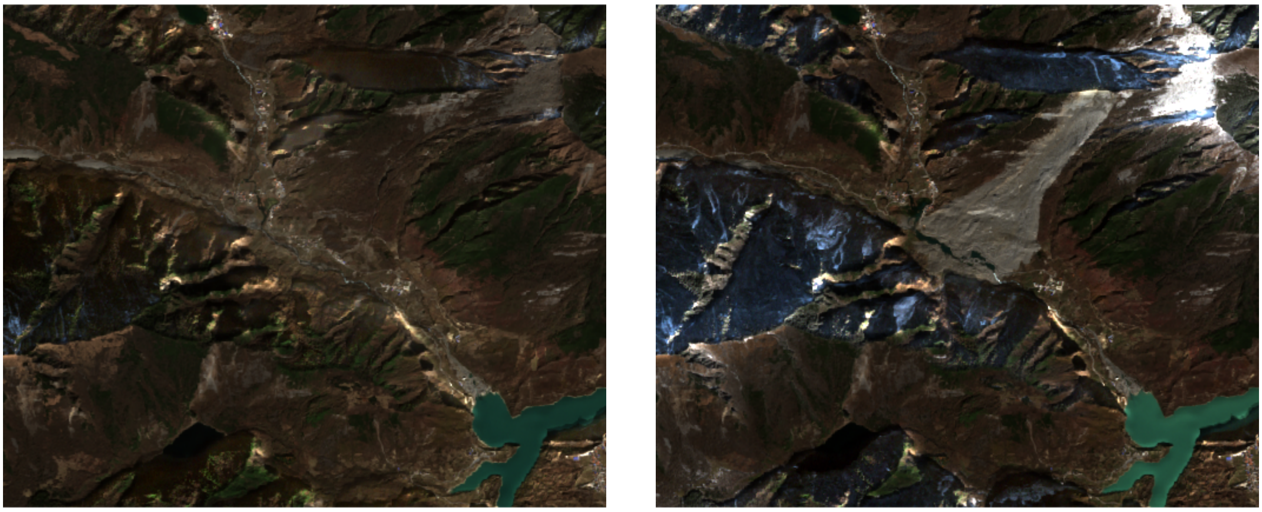


(a) Pre-event VV (b) Post-event VV (c) Difference

Figure 5: Sentinel-1 VV backscatter change (descending IW).

5 Optical Mapping and Spectral Indices

Sentinel-2 median composites (Figure 6) show the post-event debris scar replacing forest and village structures. NDVI drops sharply over the source and run-out zones (Figure 7), confirming vegetation loss. NDWI shifts along the valley floor after the event (Figure 8), consistent with altered drainage and ponded water in the debris field—similar hydrological effects noted in post-disaster optical studies of the area.



(a) Pre-event RGB (b) Post-event RGB

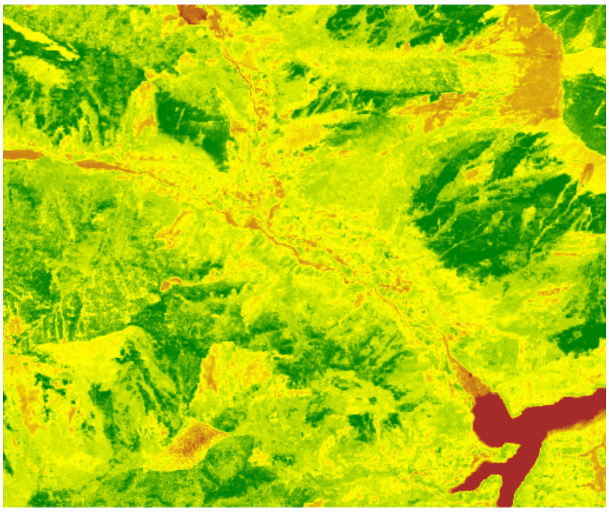
Figure 6: Sentinel-2 true-colour composites.

6 Conclusions

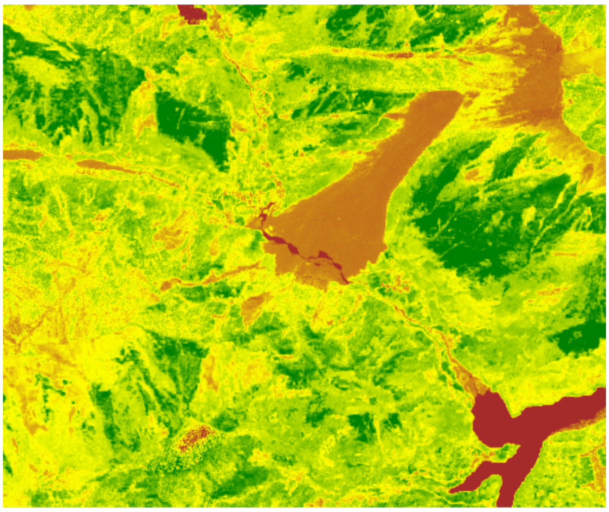
1. SBAS on LiCSAR data confirms pre-failure creep on the Xinmo source slope, in broad agreement with Intrieri *et al.* although our sparse stack and simpler processing underestimate peak rates and miss the final April–June 2017 acceleration.

2. Coherence loss does not map the landslide; VV backscatter change gives a rougher but more localised radar signal.

3. Sentinel-2 RGB, NDVI, and NDWI provide the clearest extent mapping and quantify vegetation loss and hydrological change after the event.

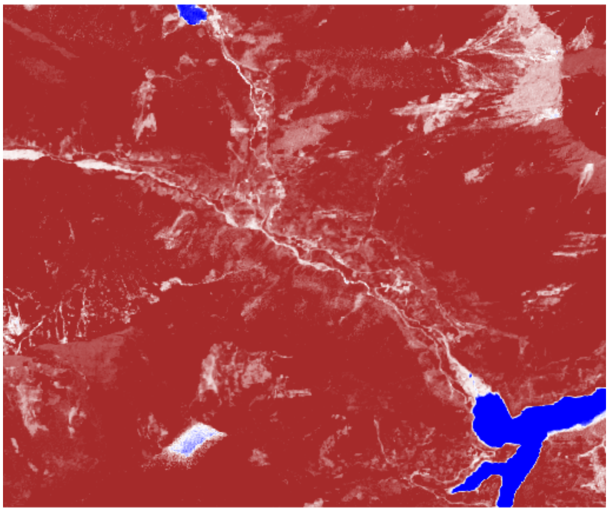


(a) Pre-event NDVI

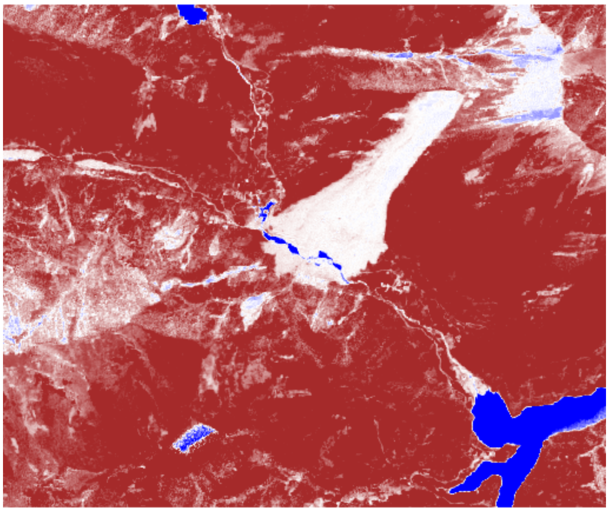


(b) Post-event NDVI

Figure 7: NDVI before and after the landslide.



(a) Pre-event NDWI



(b) Post-event NDWI

Figure 8: NDWI before and after the landslide.

References

[1] E. Intrieri *et al.*, "The Maoxian landslide as seen from space: detecting precursors of failure with Sentinel-1 data," *Landslides*, vol. 15, pp. 123–133, 2018. https://doi.org/10.1007/s10346-017-0915-7